

A. **Modular Arithmetic.** We provide a systematic way to work with remainders.

Let n be a positive integer.

We say integers a and b are **congruent modulo n** and write

$$a \equiv b \pmod{n}$$

if and only if:

So for example $-31 \equiv 11 \pmod{7}$ because:

$-31 \not\equiv 11 \pmod{8}$ because:

Let n be a positive integer, and a be an integer so $a \div n$ leaves remainder r .

Then: $a \equiv \quad \pmod{n}$

Proof. Use the division algorithm to write

$$a =$$

$$\text{Then } a - r =$$

Thus:

□

For example: $38 \equiv \quad \pmod{8}$

$$-51 \equiv \quad \pmod{8}$$

Hence every integer is congruent modulo n one of the remainders:

We call the remainders upon $\div n$ the **least residues** modulo n .

Let n be a positive integer. Two integers are congruent modulo n if and only if

Proof. Let integers a and a' be given.

Using the division algorithm we write

$$a =$$

$$a' =$$

$$\text{Then } a - a' =$$

Thus $n \mid a - a'$ if and only if:

Since $|r - r'|$ is in:

this occurs if and only if:

□

B. Properties of Modular Arithmetic. We prove a few properties.

A full list of properties is in your reference sheet.

Let n be a positive integer, and a, b, c be integers.

If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv$

Proof. Assume $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. Then

$$a - b =$$

$$b - c =$$

$$\text{So } a - c =$$

Hence:



Let n be a positive integer and a, b, c, d be integers.

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then:

- $a + c \equiv$

- $ac \equiv$

Proof. We only prove the second bullet point.

Assume $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Then

$$a - b =$$

$$c - d =$$

$$\text{Then } ac - bd =$$

Hence:



Let n be a positive integer and a, b be integers, and m be a positive integer.

If $a \equiv b \pmod{n}$ then $a^m \equiv$

Proof. Assume $a \equiv b \pmod{n}$. Then

$$a =$$

By the binomial theorem

$$a^m =$$

Observe that n necessarily divides:

$$\text{So } a^m =$$

Hence:



Example 1. Find the remainder when

$$1! + 2! + 3! + 4! + \cdots + 100!$$

is divided by 12.

Solution. Another way of describing our task is

First observe that 12 divides all summands from:

Thus working modulo 12 we have

$$1! + 2! + 3! + 4! + \cdots + 100! \equiv$$

So the remainder upon division by 12 is:

□

Example 2. Prove that 41 divides $2^{20} - 1$.

Solution. Another way of describing our task is

We start from the lefthand side and work modulo 41.

$$2^{20} \equiv$$

□